

ITT320 INTRO TO COMPUTER SECURITY

Chapter 10 : Security Policy

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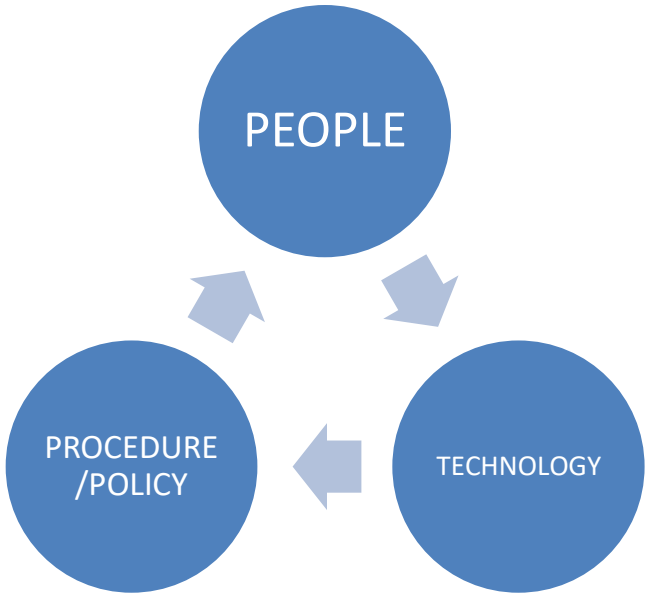
Objectives

- Create effective user policies
- Outline effective sys admin policies
- Define effective access control
- Generate effective development policies



Introduction

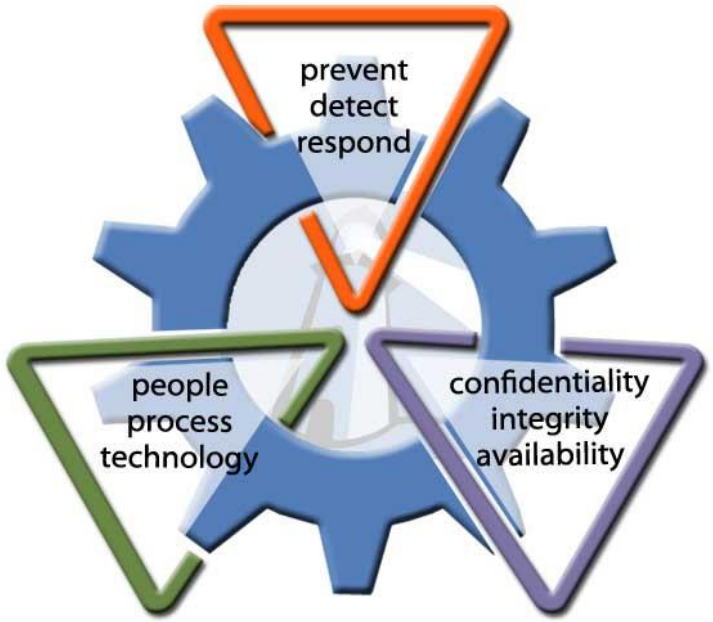
- Security technologies and all expensive perimeter devices won't guarantee 100% security and effectiveness if people do not follow appropriate procedure.
- E.g : Social Engineering and virus attack (human intervention)
- Must Remember this triangle



What is Security Policy?

- Security policy is a document that states in writing how a company plans to protect the company's physical and information technology (IT) assets.

- This policy can effectively guide you as you implement and manage security.



Cyber Security Triads

Defining User Policies

- User policies outline specifically how people may use systems and how they may not.
- The policy must be very clear and specific.
- Vague statement should be avoided that can lead to misuse such as password sharing, copying co. data etc.

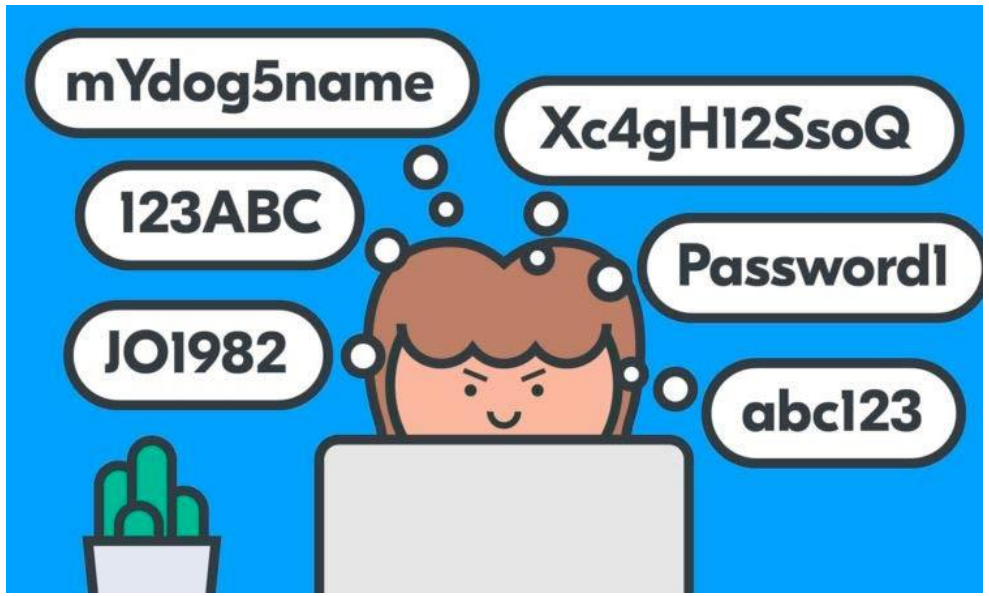


Defining User Policies

- They are several areas that effective user policies that the company must cover:

1.Passwords

- Enforce password length and minimum character
- 6-8 character long with combination of alphanumeric, numbers and symbols. E.g 123D0g@#
- Refer Previous Chapter

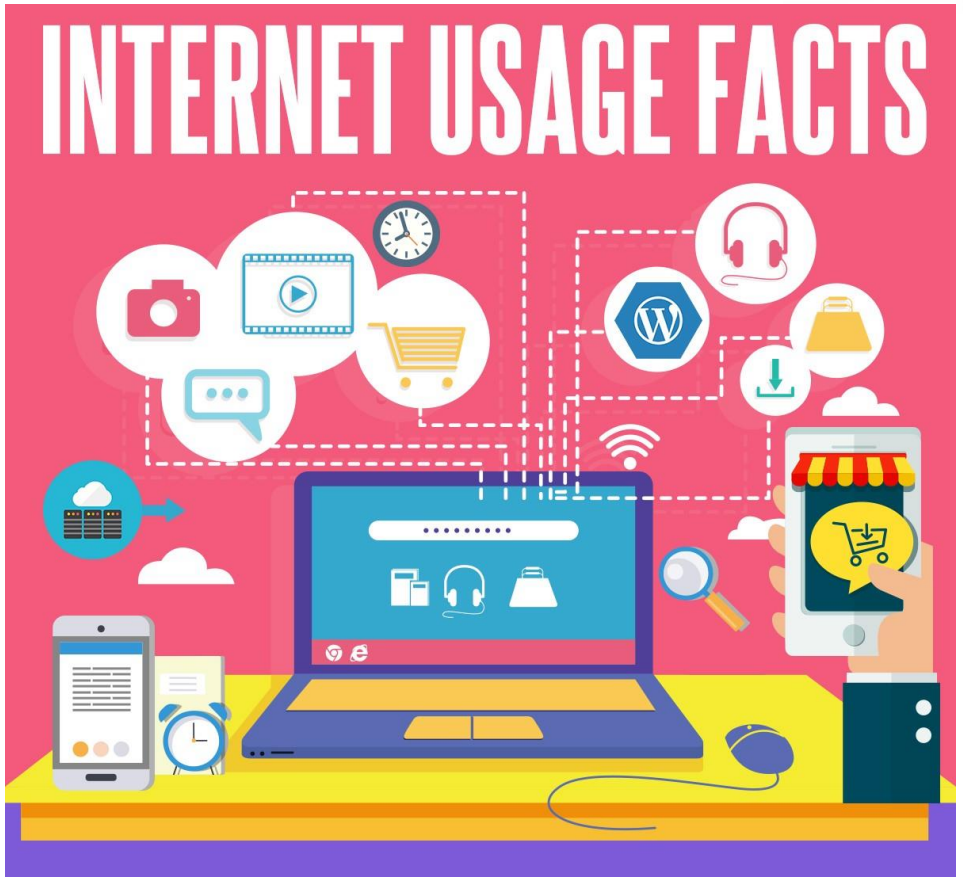


Defining User Policies

- They are several areas that effective user policies that the company must cover:

2.Internet Use

- For business / work only
- No chat rooms / Mudah.my / Website that consumes lot of traffics
- Not legitimates website



Defining User Policies

3.Email Attachments

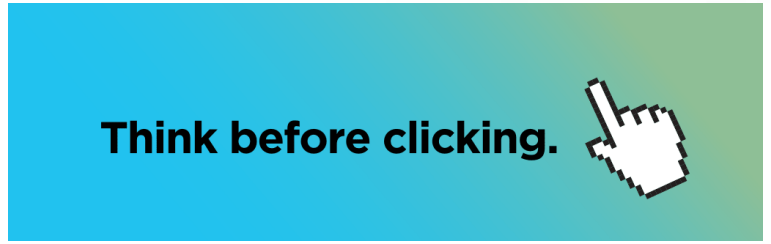
- Allow legitimate business document (.ppt, .doc, . xls)
- Disallow unknown document extension (.dat, .exe)
- Some security company disallow image files
- Limit file size to 10mb only
- Disallow email sender address which has been banned by Security company / Google



Defining User Policies

4.Software Installation and removal

- User should not allowed to install anything in company machine



5.Desktop Configuration

- User may change desktop background, font size, resolution etc. – can lead to getting virus if user keep on downloading desktop wallpaper from internet.
- E.g mypic.jpg is actually mypic.jpg.exe



Defining System Administrator Policies

Procedures that administrator need to follows for adding users, removing users, dealing with security and changing any system.



1) New Employee

   Define access, create account, job function

2) Leaving Employee

   Terminate account login asap

   Discontinue system and physical access

   Email, internet access, wireless, cell phones

Defining System Administrator Policies

3) Change Request

Form -→ Check Requirement → Make Change

Apply to IT unit, DB Change, System Change etc.



Defining System Administrator Policies

4) Security Breach – How the process in facing below security breaches:

- Virus Infection
- Denial of Services
- Intrusion by a Hacker



Read your textbook to find types of Security Breaches



Defining Access Control

- User's access to some resources must be defined specifically.
- Admin cannot allow anyone and everyone complete access to everything
- E.g Manager should be given all access to entire network system
- However, extremes in defining policies are not practical.
- Should apply the concept of Least Privilege.



SUMMARY

- User Policies
  - Password
  - Internet Uses
  - Email Attachment
  - Software Installation
  - Desktop Configuration



SUMMARY

- System Administrator Policy
  - New Employee
  - Leaving Employee
  - Change Request
  - Security Breach